

Year 9 Science
Physical Sciences
Investigation - Different Circuits

INVESTIGATION 10.4

Making changes

AIM To observe and compare the flow of electrical energy through some simple circuits

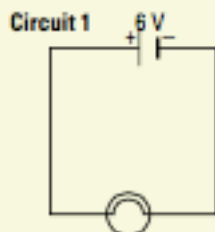
Materials:

power supply (set to 6 volts)
three 6-volt light globes and holders
6 connecting leads with alligator clips or banana plugs
piece of plastic
iron nail

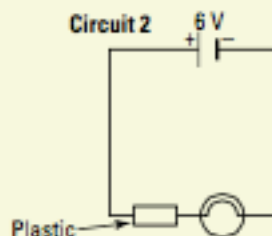
METHOD AND RESULTS

In this experiment you will be setting up each of the circuits shown in the diagrams below. Draw each circuit diagram in your workbook and record your observations of the globes next to them.

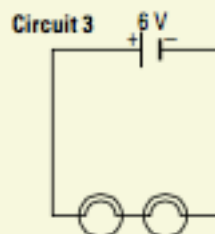
- 1 Set up circuit 1. Record the change that takes place in the globe.



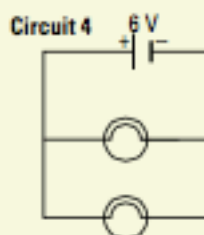
- 2 Set up circuit 2. Record the change that takes place in the globe when the plastic is replaced with an iron nail.



- 3 Set up circuit 3 so that both globes light up. The two globes are connected in series.



- 3 Are the globes glowing as brightly as the globe in circuit 1?
- 4 Predict what will happen if you disconnect the lead between the two globes in circuit 3.
- 5 Disconnect the lead and record the changes that take place in each of the globes.
- 6 Set up circuit 4 so that both globes light up. The two globes are connected in parallel.



- 6 Are the globes glowing as brightly as the globes in circuit 1?
- 7 Predict what will happen if you disconnect one of the leads between the two globes in circuit 4.
- 8 Disconnect the lead and record the changes that take place in each of the globes.

DISCUSS AND EXPLAIN

Circuit 1

- 9 Explain why the change in the globe took place.

Circuit 2

- 10 Explain the change that took place when the piece of plastic was replaced with the iron nail.

Circuit 3

- 11 Explain why the globes behaved differently from the way the globe behaved in circuit 1.
- 12 Was your prediction about what would happen when you disconnected the lead correct?
- 13 Explain the change that took place when you disconnected the lead.

Circuit 4

- 14 Was your prediction about what would happen when you disconnected the lead correct?
- 15 Explain the change that took place when you disconnected the lead.